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Functional Requirement Document (FRD)

NHCX-PMJAY-HMIS Integration

Prepared for: NHA, ABDM-enabled HMIS

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1. INTRODUCTION

This Functional Requirement Document outlines the end-to-end integration process to enable PMJAY claim submissions through any ABDM-enabled HMIS system via NHCX.

1.1 Context

Context: Current Challenges with TMS-Based PMJAY Workflow



PMJAY claims processing is currently dependent on the TMS 2.0 Provider System, irrespective of hospitals having their own Hospital Management Information Systems (HMIS).

Challenges



Duplication of data entry

Parallel data entry in hospital HMIS and TMS increases administrative workload and the risk of errors.



Lack of System Interoperability

Poor interoperability prevents seamless exchange of beneficiary, clinical, and claims data across systems.



Single-System Dependency

Exclusive reliance on a single provider system impacts scalability and operational flexibility



Restricted Innovation and Customisation

System constraints restrict innovation and flexibility for hospitals and technology partners

Solution: Transition to ABDM-Enabled HMIS Integrated with NHCX



Elimination of Mandatory TMS Dependency

Unlike the current model where PMJAY processing must be done on TMS 2.0, hospitals can operate PMJAY workflows entirely within their own HMIS without mandatory portal usage. Claims move directly from hospital systems to payer systems via NHCX.

Single-Source PMJAY Data Capture

PMJAY data is recorded once within the hospital HMIS and reused for claims processing, removing the need to re-enter or reconcile data across systems (thereby reducing errors)

Greater Choice of PMJAY-Compliant Systems

Hospitals are no longer tied to a single PMJAY processing system and can choose any ABDM-enabled HMIS

Scalable PMJAY Operations for High-Volume Hospitals

HMIS-based integrations support higher transaction volumes more reliably than portal-based workflows.

Benefits: What this means for PMJAY-empowered Hospitals



PMJAY Workflow Embedded in Hospital Operations	PMJAY cases can be handled as part of routine hospital workflows within HMIS, rather than as a separate, portal-driven activity. Hospital PMJAY operations are no longer impacted by portal downtime, access constraints, or concurrent user limitations.
Availability of Structured Data	Availability of structured data will enable automation of Pre-Auth/Claim resulting in low processing cost and time. This will reduce manual errors and improve the data quality (enabling basic requirement for use of any AI model) as well as aid in fraud control.
Improved Internal Audit and Financial Management	All PMJAY-related clinical and financial records reside within hospital systems, simplifying audits, internal reviews, and compliance. PMJAY claims data flows natively into hospital billing and accounting systems, improving reconciliation and financial reporting.
Simplified Training and User Management	Staff require training on a single system instead of managing separate user roles and workflows across HMIS and TMS.

1.2 Difference between NHCX Integration and NHCX-PMJAY Integration

Once NHCX integration is completed successfully, an integrator can send claims to any private insurer. However, as PMJAY is a national-level assurance program, there are scheme-specific modifications that need to be implemented (the APIs remain the same). This includes:

Item	Description
Utilising the Insurance Plan Response	- The PMJAY scheme extensively uses the InsurancePlan FHIR bundle, which is configured at the hospital level and serves as a key reference for scheme operations. It includes details such as available specialties, package costs for services, standard treatment guidelines, questionnaires, and mandatory documents required during pre-authorisation and claim submission. - As this configuration drives several downstream workflows, it is critical for enabling subsequent steps in the pre-authorisation and claims lifecycle. Please refer to Section 8.1 for more details.
Biometric authentication of the Beneficiary	The PMJAY scheme mandates biometric authentication of a beneficiary during registration, treatment and discharge. This is an additional API that has to be implemented. Please refer to section 8.2 for more details.

Structured data exchange	For a HMIS to process PMJAY claims, it is mandatory to send supporting health information in ABDM-defined structured Health Information Types. Please refer to sections 8.5.0 or 8.6.0 for more information on the same.
Query flow	The PMJAY payer does not use the Communication API for queries. Instead, a query is raised by the payer with the relevant workflow ID for preauth/claim which has to be responded to by the provider with the relevant workflow ID (preauth/claim bundle structure remains the same). Please refer to sections 8.4.4 or 8.5.2 for more information.

API Flows

NHCX normal flow	NHCX-PMJAY-HMIS flow
Get Insurance Plan	Same
Get Policy	Same
Coverage eligibility check	Same
Pre-Auth	Same (but before pre-auth need to do mandatory biometric authentication of beneficiary)
Communication Request	Communication API is not used. Instead, a query is raised by the payer with the relevant workflow ID for preauth/claim which has to be responded to by the provider with the relevant workflow ID.
Claim Submission	Same
Payment Notice	Same
Status Check	Same

1.4 Scope

This FRD covers:

- NHCX workflow adoption
- Standardised FHIR-based claim requests
- Digital documentation exchange
- Integration with existing PMJAY modules (TMS)

- Security, audit trail, and compliance

1.5 Target Audience

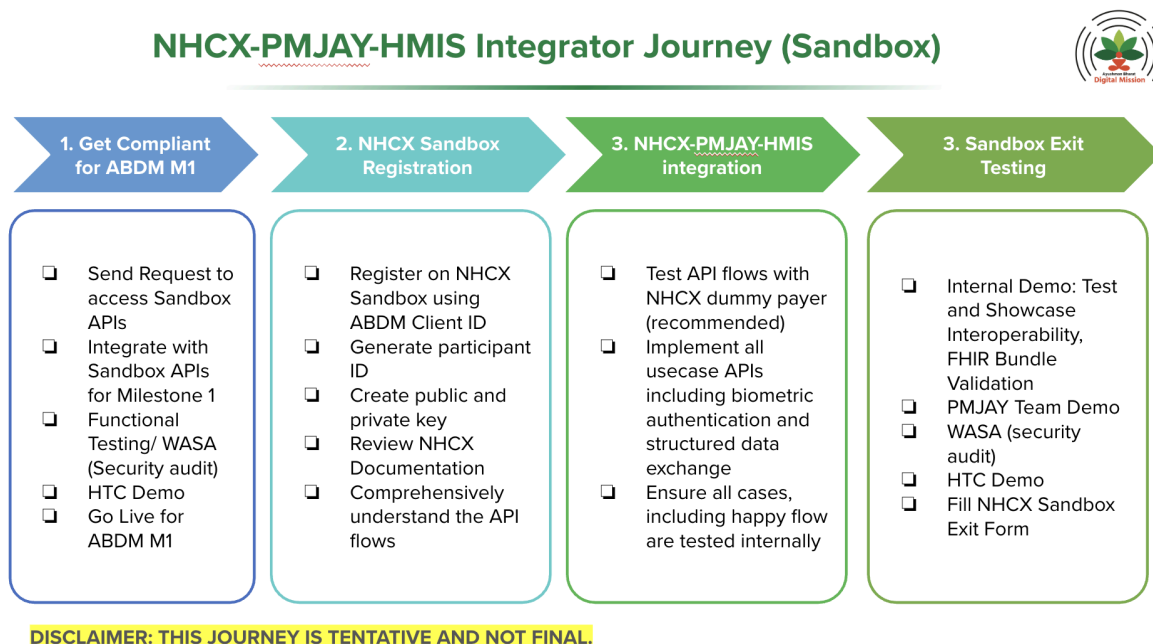
- National Health Authority
- State Health Agencies
- Insurance Companies / TPAs
- Hospital Providers
- IT Vendors

1.6 References

- ABDM – NDHM Standards
- NHCX API Specifications
- PMJAY Scheme Guidelines

1.7 Sandbox Journey for NHCX-PMJAY Integration

This document describes processes specific to processing PMJAY claims via NHCX. It is recommended for an ABDM-enabled HMIS to build a strong understanding of NHCX and NHCX API Flows before initiating PMJAY-integration.



Please note:

- (1) To commence the NHCX-PMJAY-HMIS integration, the integrator has to share the Participant ID, Client ID and Registry ID with the NHCX team so the participant can be onboarded on the staging environment for PMJAY.

- The Registry ID used during participant creation has to be used as the 'Provider ID' on sandbox.
 - A sample member ID will also be for testing purposes.
- (2) Upon successful completion of NHCX-PMJAY-HMIS Integration, the integrator will receive production keys for NHCX that can then be used to process claims for private insurers as well.

The following technical documents can be referred to:

- [ABDM Documentation](#)
- [NHCX Documentation](#)
- [Onboarding providers and payers in Sandbox](#)
- [Steps to generate key](#)
- [NHCX Provider Side Use Cases- Sandbox Exit Process](#)

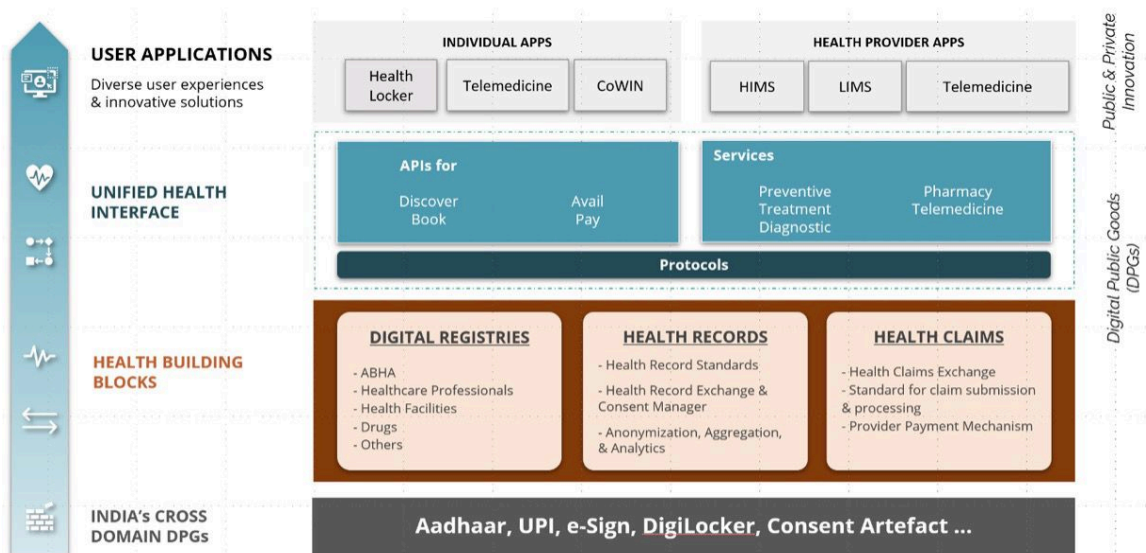
1.8 Test Cases for Exit Process

Please refer to the document titled NHCX-PMJAY-HMIS Test Cases.

2. ABDM & NHCX OVERVIEW

2.1 ABDM Ecosystem

The Ayushman Bharat Digital Mission (ABDM) aims to develop the backbone necessary to support the integrated digital health infrastructure of the country. It will bridge the existing gap amongst different stakeholders of the Healthcare ecosystem through digital highways.



For more details please visit the ABDM website: <https://abdm.gov.in/>

2.2 About the National Health Claims Exchange (NHCX)

The National Health Claims Exchange (NHCX) is developed by NHA, in consultation with Insurance Regulatory and Development Authority of India (IRDAI). With the primary purpose to

streamline and standardise the processing of health insurance claims across the country, leveraging the Fast Healthcare Interoperability Resources (FHIR) standards to ensure interoperability across diverse systems. NHCX Specification is a standardized communication protocol designed to facilitate the seamless exchange of health claim information among payers, providers, beneficiaries, and other relevant entities.

NHCX Specifications provide a comprehensive blueprint for every aspect of the envisioned claims network. They are designed to ensure technology and vendor neutrality, adaptability to evolving needs, and to foster innovation and inclusivity within the healthcare ecosystem.

For more details please visit the NHCX website: <https://nhcx.abdm.gov.in/#/>

3. PMJAY OVERVIEW

Ayushman Bharat is the Pradhan Mantri Jan Arogya Yojana or PMJAY as it is popularly known was launched by the Hon'ble Prime Minister of India, Shri Narendra Modi on 23rd September 2018 in Ranchi, Jharkhand. PMJAY is the largest health assurance scheme in the world which aims at providing a health cover of Rs. 5 lakhs per family per year for secondary and tertiary care hospitalization to over 10.74 crores poor and vulnerable families (approximately 50 crore beneficiaries) that form the bottom 40% of the Indian population. PMJAY is fully funded by the Government and cost of implementation is shared between the Central and State Governments.

In PMJAY, claim adjudication is done through integrated workflows between three key systems –

1. Beneficiary Identification System (BIS),
2. Transaction Management System (TMS) and
3. Hospital Empanelment Module (HEM)

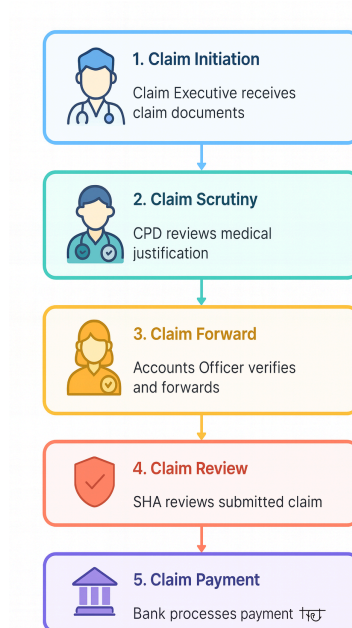
The key tasks are performed under Transaction Management System (TMS), partially at the time of Pre-authorization by Pre-authorization Processing Doctor (PPD) and later at the time of final claim settlement by Claim Processing Doctor (CPD) based on the documents received from the hospital.

4. EXISTING PMJAY WORKFLOW

4.1 Claims Adjudication Process

Claims adjudication refers to the decision on two key aspects of a claim: whether the claim is admissible under the terms of policy/Scheme and if yes, what is the quantum payable. It applies to the final decision on claims payment. The decision involves cross verification of all-important aspects – covered person, medical condition – symptoms, diagnosis, treatment, policy exclusions, period, available sum insured, pre-agreed tariff/package rate, empanelled hospital etc.

The Process is illustrated below:



4.2 Claims Payment Process

Post approval of a claim by SHA (Trust model) or SHA – IC (Insurance model), claim payment would be initiated by the bank. Amount will be transferred to EHCP (Empanelled Healthcare Provider) account. SHA will follow applicable PFMS guidelines for making claims payment.

Users (Roles) involved:

- **MEDCO** – Medical Committee
- **CEX** – Claim Executive
- **CPD** – Claim Processing Doctor
- **ACO** – Accounts Officer
- **SHA** – State Health Authority

The detailed workflow along with Users(roles) is as below:

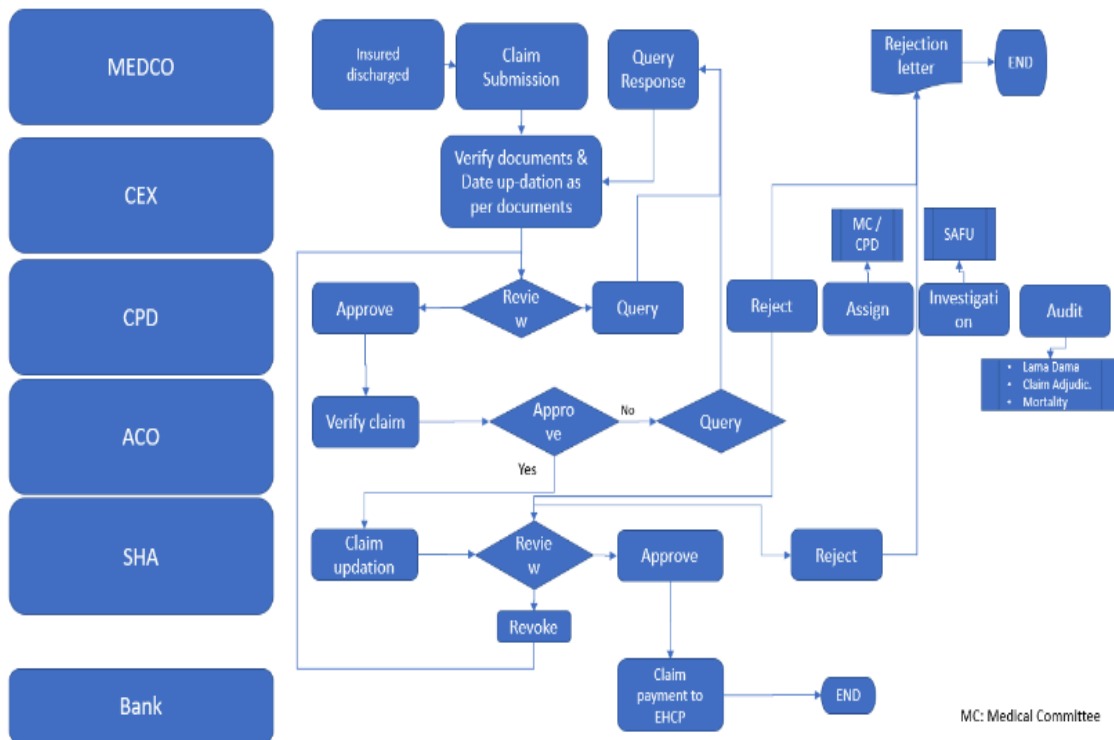


Figure Claims 1 Claims Process Flow

5. Transaction Management System (TMS) for Providers

5.1 About TMS Provider Application

TMS Provider Application was designed for the hospitals to encompass the entire lifecycle and engagement with PMJAY beneficiaries and shall have modules/ functionalities like Patient Registration, Patient Pre-Authorization, Treatment, Patient Discharge, Claims, Payments, MIS Reports and Dashboards. Any Hospital Management Information System (HMIS) or Digital Solution Company (DSC) that aims to process PMJAY claims should have the same capabilities.

5.2 Detailed End-to-end Process Flow in TMS Provider Application

Functional step	Details
Beneficiary registration and authentication	• 'Search the beneficiary' using Ayushman ID/Mobile number/ABHA and select the relevant payer • Member Card of the beneficiary is displayed • Beneficiary needs to be verified against Aadhaar using fingerprint/Iris/face auth to proceed • Proceed without Aadhaar option available for exceptional circumstances

	○ A template undertaking document is required for the beneficiaries who do not have an Aadhaar card with them but have the PMJAY card. This will have to be submitted at the time of claim submission to the payer. ● Communication address, Care Plan, Patient Attendant details needs to be filled ● Authentication Consent to be uploaded ○ To be signed by the doctor and patient. This has a specific format that providers should have access to. There is a fraud engine that detects if this is the right document and it passes only if it is valid. ● Once all mandatory details have been filled and submitted, patient is registered and wallet balance is displayed
Pre-authorisation	● Following information will have to be fetched from the HMIS: ○ Medical information (which includes general findings, personal history & family history) ○ Admission information (which includes authentication consent & admission details) ○ Treatment (which includes diagnosis (ICD codes), treatment plan, investigations & care team details) ○ Finance details ● Once the benefits are added, there is a check which verifies if there is adequate balance to cover the benefits ● The pre-auth form is then validated ● After the pre-auth has been initiated, the request will go to PPD (preauth processing doctor) at payer end for further approval. If the selected procedure is an auto approved procedure, it can automatically move for treatment.
Discharge details	● Once the patient is ready for discharge, provision to select the type of discharge as normal/LAMA/DAMA/Death. Details to be captured vary based on the type of discharge. ● Documents need to be fetched from the HMIS based on the type ● Download the Mangalkamna Patra and handover to the beneficiary and upload the same in the feedback form. ● Patient needs to be verified using biometric (proceed without aadhaar option available)
Claim initiation	● Finance section details to be filled where the amount and incentive details and amount claimed as per hospital bill needs to be verified and need to upload/capture the supporting documents say., hospital bill number, date,

	hospital bill and any other related documents • Provision to upload post OP investigations
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Note: (1) E-Rupi initiation form is not required; (2) For more details and edge cases, please refer to the [TMS Provider User Manual](#).

6. FUNCTIONAL REQUIREMENTS

6.1 High-Level NHCX Process Flow

Step	NHCX API	Action/requirement	Frequency
Pre-condition	InsurancePlan	Retrieve policy details (required to auto-fill benefits, validate admissibility, STGs, documents, latest treatment packages, limits, etc.) Update InsurancePlan data for active beneficiaries	Once per policy, or weekly
Beneficiary registration and authentication	Coverage Eligibility Check	Verify patient eligibility	Per registration
	Biometric authentication	Verify patient identity	Per registration
Pre-authorisation	Insurance Plan	Treatment, STG questionnaire, mandatory documents and benefit options fetched directly the payer system	Per claim
	Pre-auth	Submit as per InsurancePlan references, directly fetched from the HMIS.	
Discharge	InsurancePlan	Documents required at the time of discharge to be pre-populated	Per claim
Claim submission	InsurancePlan	Treatment, STG questionnaire, mandatory documents and benefit options fetched directly the payer system	Per claim

	Claim	Submit as per InsurancePlan references, directly fetched from the HMIS.	
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7. INTEGRATION REQUIREMENTS

7.1 API Requirements

- All communication via **NHCX Gateway APIs**
- JSON + FHIR R4 formats
- Standard HTTP status codes

7.2 Authentication

- JWT tokens
- Digital Signing Certificate required

7.3 Error Handling

Exception handling is a critical aspect of building robust and user-friendly REST APIs. It helps to

achieve the following:

1. **Improved User Experience:** Clients receive clear and meaningful error messages instead of cryptic stack traces.
2. **Debugging and Maintenance:** Proper exception handling makes it easier to identify and fix issues.
3. **Consistency:** Standardized error responses ensure consistency across your API.

Best Practices for Exception Handling

1. **Use Specific Exceptions:** Create custom exceptions for specific error scenarios (e.g., `ResourceNotFoundException`, `ValidationException`).
2. **Return Meaningful Error Messages:** Provide clear and actionable error messages to the client.
3. **Log Exceptions:** Log exceptions for debugging and monitoring purposes.
4. **Use HTTP Status Codes Correctly:** Follow REST conventions for status codes (e.g., 404 for missing resources, 400 for bad requests).
5. **Avoid Exposing Sensitive Information:** Do not include stack traces or internal details in production error responses.

System must support:

- NHCX error codes

- Retry logic
- Human-readable messages

For more information please refer to the following documents: [API Response Handling to avoid Failures](#), [Standard error codes](#)

7.4 Data Structure

All data objects must be encrypted and transmitted within the API request body, inaccessible to NHCX gateways. Fast Healthcare Interoperability Resource (FHIR) Bundle resource has been adopted for encapsulation of the data shared across NHCX APIs. A Bundle can carry a collection of resources meeting certain criteria as part of any service operation. The structure of the e-claim Bundle should follow certain guidelines. These bundles should contain FHIR bundles of the "collection" type, where the type of bundle is specified by "bundle.type".

For more information please refer to the following documents: [Implementation Guide for Adoption of FHIR in ABDM and NHCX](#), [NHCX Profiles - FHIR Implementation Guide for ABDM v6.5.0](#), [Benefit Category - FHIR Implementation Guide for ABDM v7.0.0](#)

8. API Specifications

8.1. Insurance Plan

The **InsurancePlan** is a structured digital representation of the policy under which a patient is covered in the PMJAY scheme. Since PMJAY is strictly package-based (bundled costs for each service), referencing the correct InsurancePlan is critical for:

- Ensuring only admissible services/benefits are claimed.
- Avoiding claim rejections due to non-compliance with scheme guidelines.
- Providing transparency and auditability for both providers and payers (State Health Agencies/Insurance Companies).

Key input parameters	• Provider ID • Policy Code • Participant ID
Key value sets returned	• Only has the list of specialties relevant to each provider (response is based on the Provider ID sent) • List of covered services/benefits and their limits. • Mandatory documents as per guidelines for both pre-authorisation and claim submission. This document list will be available both for the policy-level and for benefit-specific requirements as well. • Standard Treatment Guidelines (STGs) and clinical protocols. • Policy conditions, exclusions, and renewal information.

Technical Implementation Guidelines	• The size of the plan object is more than 20 MB sometimes and the system should be capable of receiving the payload and storing the details in a structured, queryable format linked to each policy record. • Updating Insurance Plan internally: - o Run weekly API calls to refresh the InsurancePlan for all active beneficiaries to ensure the latest treatment packages, benefits, and policy rules are applied. - o Trigger an immediate refresh whenever a policy is renewed or amended by the payer. - o Implement version control for tariff and package updates, as outdated tariff versions will cause rate mismatches and automatic claim rejection due to suspected tampering. - o Maintain detailed audit logs of every InsurancePlan fetch, refresh, version update, and the version used during preauth or claim submission for compliance and traceability. • Ensure all required documents and Standard Treatment Guideline (STG) checklists are attached as per InsurancePlan requirements. STG is mandatory for PMJAY and it should be rendered dynamically as per the treatment plan selected. - o Under insurance plan - (provider gets url in the questionnaire) - under questionnaire response this url needs to be sent in preauth/claim - o Description of questionnaire - Previously item description was sent in the prefix parameter now it will go as a text in insurance plan • Notify users if there are changes in policy guidelines or if a selected package/service is no longer covered.
APIs to be called	/v1/insuranceplan/request
Callback API to be Implemented	/v1/insuranceplan/on_request
Use case description	Request insurance plan details from the Payer via HCX
Protocol status	request.initiated
Request Payload	Encrypted payload of TaskBundle
Validations	Payload should be validated against the profiles published by NRCES.

The entity to implement the API is called	Payer
Callback API logic	Callback API should be implemented by provider systems. It should accept the payload in two forms, and it will be derived based on the “type” parameter of the response. 1. Encrypted format (As per RFC7516) of the payload InsurancePlanBundle using the private key of the provider entity. Encrypted payload comes as a response only when the payer has processed the request and responded, when everything is validated and verified. 2. ProtocolResponse. ProtocolResponse comes as a response only when the payer is able to process the request because the payload is invalid or not.
Sample request and response	• PMJAY-insuranceplan request sample.txt • PMJAY-insuranceplan response sample.txt
Other reference documents	• InsurancePlan - FHIR Implementation Guide for ABDM v6.5.0 • Sample IDs ○ Member ID: Will be assigned by the Integration during onboarding of the integrator ○ Sample payer id: 1518@hcx ○ Sample policy code: PMJAY/CH/S/G ○ Sample HFR: currently, registry ID is mapped for stage and test environment

Using InsurancePlan in Claims Submission

Step	Details
Fetching and storing the insurance plan	• The Insurance Plan should be fetched at the hospital-level based on the specialties and treatments a hospital is authorised for under the PMJAY scheme. • This call should fetch the latest value sets: covered benefits, limits, required documents, STGs, and package details. • This should be stored locally. • The InsurancePlan API call is typically required once per policy period (e.g., at renewal), but it is recommended to refresh this data at least weekly to capture any updates in policy guidelines.

Patient Registration and Coverage Eligibility	- When a patient presents at a provider facility, the system must first perform a Coverage Eligibility Check via the NHCX API. - Upon confirmation of eligibility, the patient is registered under the PMJAY scheme in the provider's HMIS.
Fetching InsurancePlan Details	The provider system must fetch the locally stored InsurancePlan to retrieve the policy details for the patient based on the policy code received from coverage eligibility request (all policies relevant to the hospital would be pre-loaded in your system so you would only be fetching it)
Populating Benefits and Admissible Checks	- The provider system should automatically populate the preauth/claim form with benefits and limits as per the InsurancePlan. - Admissible checks must be performed to ensure: - Only eligible packages/services are selected. - All mandatory documents and clinical data (as per STGs) are attached. - Any package-specific conditions or exclusions are validated before submission
Preauthorisation and Claim Submission	- Before submitting a preauthorization or claim, the system should: - Fetch all authorization requirements using CoverageEligibilityRequest through NHCX APIs. - Validate that all InsurancePlan requirements (documents, clinical parameters, package selection) are met. - Submit the preauthorization/claim with references to the InsurancePlan and associated value sets.

8.2 Biometric Authentication

This is not a NHCX API and has been created specifically for the PMJAY payer. To comply with the mandatory requirements of the PMJAY scheme, all participating hospitals must follow the below authentication protocol during patient registration, the pre-authorization and discharge workflow. Please refer to the Postman Collection.

Biometric Authentication Using ABHA APIs	- At the time of registering the patient or initiating a pre-authorisation request, hospitals shall perform biometric authentication of the beneficiary using fingerprint, iris or face auth to ensure the physical presence of the patient. - Upon successful biometric verification, the system shall generate a User Token.
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	• This User Token must be passed as a header parameter for all subsequent PMJAY claim-related events, including Coverage Eligibility Check, Pre-Authorization Request Submission or any additional PMJAY claim events requiring proof of beneficiary presence. This token will be validated at payer systems before considering it as legitimate request. • Biometric authentication is also required at the discharge stage. This token has to be passed for Claim Submission.
Token Validity and Refresh Mechanism	• The User Token issued after authentication remains valid for 30 minutes. • Hospital systems must ensure that the token is refreshed automatically, as required, until completion of the relevant transaction cycle. • If the token expires at any stage, a fresh biometric authentication must be initiated. This step is mandatory to meet the PMJAY requirement of ensuring proof of patient presence at the hospital. Please refer to API specifications and the postman collection for more details.
Applicability	• This authentication process is applicable only for beneficiaries whose ABHA ID is linked with their PMJAY card. • For beneficiaries without ABHA linkage, the alternate PMJAY-approved KYC protocols may be followed as per the existing guidelines.
Handle Exemption	• In cases where biometric/aadhaar authentication is not feasible, providers must obtain an Aadhaar exemption consent document. This document must be signed by both the patient and the hospital representative, digitally stored, and linked to the beneficiary record to maintain compliance with ABDM standards.
Key points	• Biometric authentication or a valid exemption is mandatory for provider systems to submit claims in accordance with PMJAY policy. • The process ensures compliance, security, and verifiable beneficiary identification throughout the claim submission workflow.

8.2.1 API Details

1. Auth INIT

Request curl:

```
curl --location --request POST 'https://apisbx.abdm.gov.in/hcx/abha/biometric/auth/init' \
--header 'accept: */*' \
--header 'Content-Type: application/json' \
--header 'Authorization: Bearer {{Token}}' \
--header 'process: PreauthlDischarge' \
--header 'payerid: 123@hcx' \
--data-raw '{
  "scope": ["abha-login", "aadhaar-bio-verify"],//["abha-login", "aadhaar-bio-verify"] or
["abha-login", "aadhaar-face-verify"] or ["abha-login", "aadhaar-iris-verify"]
  "loginHint": "abha-number",
  "loginId": "91-XXXX-XXXX-1234",
  "otpSystem": "aadhaar",
  "authMode": "FINGERPRINT"//FINGERPRINT,IRIS, FACE_AUTH,
}'
```

Response:

```
{
  "txnId": "8c8a12e3-xxxx-4278-xxxx-10acffa44f07",
  "authMode": null,
  "message": "FingerPrint authentication request successfully sent.",
  "status": null
}
```

2. Auth Verify

Request curl:

```
curl --location --request POST
'https://apisbx.abdm.gov.in/hcx/abha/biometric/auth/verify' \
--header 'accept: */* ' \
--header 'Content-Type: application/json' \
--header 'Authorization: Bearer {{Token}}' \
--header 'process: PreauthIDischarge' \
--header 'payerid: 123@hcx' \

--data-raw '{
  "scope": ["abha-login",
"aadhaar-bio-verify"],/["abha-login", "aadhaar-bio-verify"] or ["abha-login", "aadhaar-fac
e-verify"] or ["abha-login", "aadhaar-iris-verify"]

  "authData": {
    "authMethods": [
      "bio"//bio | iris |face
    ],
    "bio": {
      "txnId": "d21b3db9-xxxx-xxxx-8eed-8f75e7f86b9f",
      "fingerPrintAuthPid": "string"
    },
    "face": {
      "txnId": "d21b3db9-xxxx-xxxx-8eed-8f75e7f86b9f",
      "faceAuthPid": "string"
    },
    "iris": {
      "txnId": "d21b3db9-xxxx-xxxx-8eed-8f75e7f86b9f",
      "irisAuthPid": "string"
    }
  }
}
```

```
},
"otp": {
  "txnId": "d21b3db9-xxxx-xxxx-8eed-8f75e7f86b9f",
  "otpValue": "123456"
}
},
"authMode": "FINGERPRINT" //FINGERPRINT,IRIS, FACE_AUTH
}'
```

Response:

```
{
  "txnId": "d21b3db9-478a-xxxx-xxxx-8f75e7f86b9f",
  "authResult": "success",
  "message": "OTP verified successfully",
  "token": "eyJZhx....",
  "refreshToken": "eyJZhx....",
  "expiresIn": 1800,
  "refreshExpiresIn": 1296000,
  "accounts": [
    {
      "ABHANumber": "91-XXXX-XXXX-1234",
      "preferredAbhaAddress": "91XXXXXXXXX1234@sbx",
      "name": "Test Name",
      "gender": null,
      "dob": null,
      "verifiedStatus": null,
      "verificationType": null,
    }
  ]
}
```

```

        "status": "ACTIVE",
        "profilePhoto": "base 64 image"
    }
]
}

```

3. Auth Refresh Token:

During any ABHA transaction, when user enrolls or logs in, an X-Auth Token is generated. This token is used to fetch the user's profile details and is valid for 30 minutes.

Along with the X-Auth Token, a Refresh Token is also provided. This Refresh Token is used to generate a new X-Auth Token through the given API. The Refresh Token is valid for 15 days.

If you need a new X-Auth Token after 15 days, you can call the same API again. It will give you a new Refresh Token, which will then be valid for another 15 days from the time it was generated.

To keep the Refresh Token active continuously, you can call the API once within every 10 days and store the new Refresh Token. This way, you can extend access to the X-Auth Token for as long as needed.

Request curl:

```

curl --location --request GET
'https://apisbx.abdm.gov.in/hcx/abha/biometric/auth/refresh/token' \

```

```

--header 'R-token: Bearer {{RefreshToken}}' \

```

```

--header 'Authorization: Bearer {{Auth}}' \

```

```

--header 'payerid: 123@hcx' \

```

```

--header 'process: PreauthIDischarge' \

```

Response:

```

{

```

```

"txnId": "d21b3db9-478a-xxxx-xxxx-8f75e7f86b9f",
"authResult": "success",
"message": "OTP verified successfully",
"token": "eyJZhx....",
"refreshToken": "eyJZhx....",
"expiresIn": 1800,
"refreshExpiresIn": 1296000,
"accounts": [
  {
    "ABHANumber": "91-XXXX-XXXX-1234",
    "preferredAbhaAddress": "91XXXXXXXXX1234@sbx",
    "name": "Test Name",
    "gender": null,
    "dob": null,
    "verifiedStatus": null,
    "verificationType": null,
    "status": "ACTIVE",
    "profilePhoto": "base 64 image"
  }
]
}

```

8.3 Coverage Eligibility

The PMJAY Coverage Eligibility Check via NHCX enables healthcare providers to verify a patient's eligibility and available benefits under the PMJAY scheme before registration or treatment. This process ensures that only eligible beneficiaries receive scheme benefits, prevent misuse, and streamline patient onboarding.

Key value sets returned	Value sets are returned depending on the following purposes: ● Discovery ● Validation
--------------------------------	--

	• Auth requirements • Benefits
Technical Implementation Guidelines	• Integrate and test the CoverageEligibilityCheck API. • Implement a callback endpoint for asynchronous responses. • Parse and validate benefit limits and coverage status. • Support family coverage validation. • Capture and store Aadhaar exemption consent documents. • Alert provider and patient if coverage is insufficient. • Register patients only after validating coverage. • Call coverage eligibility every time a new/additional treatment (should be treated as a validation) is added to ensure it is within the limits before submitting a pre-authorisation.
Example scenarios	• Individual Coverage Valid: Patient has sufficient coverage; registration proceeds. • Family Coverage with Shared Limit: Remaining family coverage is checked before registration. • Limit Exhausted: Registration is blocked if the family's limit is exhausted. • Aadhaar Exemption: Consent is captured and eligibility check proceeds if Aadhaar is unavailable.
APIs to be called	/v1/coverageeligibility/check
Callback API to be Implemented	/v1/coverageeligibility/on_check
The entity to implement the API is called	Payer
Protocol status	request.initiated
Request Payload	Encrypted payload of CoverageEligibilityRequestBundle
Validations	Payload should be validated against the profiles published by NRCES.
Callback API logic	The callback API should be implemented by provider /PHR systems. It should accept the payload in two forms, and it will be derived based on the "type" parameter of the response.

	1. Encrypted format (As per RFC7516) and decrypted using the private key of the provider entity. Encrypted payload comes as a response only when the payer has processed the request and responded, when everything is validated and verified. 2. ProtocolResponse. ProtocolResponse comes as a response only when the payer is able to process the request due to the payload being invalid or not able to decrypt, or any protocol errors.
References	• Coverage Eligibility API • CoverageEligibilityRequest - FHIR Implementation Guide for ABDM v6.5.0 • CoverageEligibilityResponse - FHIR Implementation Guide for ABDM v6.5.0

Coverage Eligibility API Request and Response based on purpose

8.3.1 Validation

Description	A check that the specified coverages are in-force is requested (and wallet balance as well)
Mandatory input parameters	• Beneficiary ID • Coverage/Plan Code • Payer ID • Provider ID
Key value sets returned	• Coverage component ◦ Period • Coverage eligibility response component ◦ Status ◦ Requester ◦ Patient detail ◦ Insurer ◦ Insurance component ■ Item • Benefit component (number of entries here will be the same as number of wallets applicable to the beneficiary) ◦ Allowed (balance) ◦ Used (consumed)

8.3.2 Discovery

Description	The insurer is requested to report on any coverages which they are aware of in addition to any specified. It provides a list of all active coverages for the beneficiary.
--------------------	---

Mandatory input parameters	● Beneficiary ID ● Coverage/Plan Code ● Payer ID ● Provider ID
Key value sets returned	● Coverage component ○ Period ● Coverage eligibility response component ○ Status ○ Requester ○ Patient detail ○ Insurer ○ Insurance component

8.3.3 Auth requirements

Description	The prior authorization requirements for the listed, or discovered if specified, coverages for the categories of service and/or specified billing codes are requested.
Mandatory input parameters	● Beneficiary ID ● Coverage/Plan Code ● Payer ID ● Provider ID ● Procedure/Package Codes
Key value sets returned	● Coverage component ○ Period ● Coverage eligibility response component ○ Status ○ Requester ○ Patient detail ○ Insurer ○ Insurance component ■ Item ● Benefit component (this will have procedure-wise auth requirement details)

8.3.4 Benefits

Description	The plan benefits and optionally benefits consumed for the listed, or discovered if specified, coverages are requested.
Mandatory input parameters	● Beneficiary ID ● Coverage/Plan Code ● Payer ID ● Provider ID ● Procedure/Package Codes
Key value sets	● Coverage component

returned	○ Period ● Coverage eligibility response component ○ Status ○ Requester ○ Patient detail ○ Insurer ○ Insurance component ■ Item ● Benefit component (this will have procedure-wise auth requirement details)
-----------------	--

Using CoverageEligibility in Claims Submission

Step	Detail
Initiate Coverage	● The provider's system sends a request to the Payer system through NHCX using the CoverageEligibilityCheck API.
Eligibility Check	● Required details: beneficiary's Payer ID, ABHA Id, PMJAY Id, Aadhaar number, or other identifiers which are supported by PMJAY.
Receive CoverageEligibilityResponse	● The Payer system processes the request and sends an asynchronous response to the provider's system through NHCX. ● The response includes beneficiary details, coverage status, benefit limits, and family coverage information.
Validate Coverage and Limits	● The provider's system parses the response to extract benefit limits and coverage status. ● It checks if the requested services/packages are covered and within the available limits. ● If coverage is insufficient or the service is not covered, the system alerts the provider and patient.
Register Patient with Validated Coverage	● If coverage is confirmed, the patient is registered for treatment. ● If Aadhaar authentication is not possible, the system captures Aadhaar exemption consent and stores the documentation.
Inform Patient and Prepare for Next Steps	● The provider communicates the coverage status and limits to the patient. ● If coverage is insufficient, alternative options are discussed.

8.4. Pre-authorisation

8.4.0 Structured Data Exchange

Supporting documents need to be sent in a fully structured FHIR format. This will have to be aligned with the Health Information (HI) Types defined under ABDM.

- When **submitting structured clinical data** as part of a **Preauthorization or Claim**, the information must be shared through the **SupportingInfo → DocumentReference** resource.
- Within the **DocumentReference.content.attachment** element, the **attachment.data** field should contain a **Base64-encoded FHIR Bundle**. This bundle should include the relevant structured FHIR resources (for example, **DiagnosticReport**, **DischargeSummary**, **WellnessRecord**, and any associated resources) required to support the preauth or claim.
- This differs from **unstructured submissions**, where the attachment typically contains a **Base64-encoded PDF or JPG document**.
- For **structured submissions**:
 - The attachment represents a **FHIR Bundle encoded in Base64**, not a static document.
 - The bundle contains the necessary **FHIR resources describing the clinical information** relevant to the claim or preauthorization.
The **DocumentReference.content.attachment.contentType** should be set to either **application/json** or **application/fhir+json**.
- For structured data, category code of supporting info needs to be among DIA, HDS, CD, INF and the value needs to be reference.
- For unstructured data, the category code of supporting info needs to be among POI, POA, DOB, DEF, FIR, ATT and the value needs to be attachment.
- Reference documents:
 - [ABDM Sandbox Health Record Formats](#)
 - [Understanding the Structure of ABDM HI Types its Creation and Validation](#)
 - [FHIR Profiles for ABDM Health Data Interchange](#)

Key input parameters	Documents to be submitted and questionnaires for Pre-Auth to be fetched from 'Coverage Eligibility API' with purpose 'auth requirements' The following dates need to be sent mandatorily in the NRCeS prescribed date-time format with the correct codes: Registration Date • Code: EDT • Display: EncounterDateTime • Code System: ndhm-supportinginfo-code
-----------------------------	--

	• Category Code: OTH • Category Display: Other • Category System: ndhm-supportinginfo-category Admission Date • Code: ADDED • Display: Admission date – Discharge date • Code System: ndhm-supportinginfo-code • Category Code: ADMD • Category Display: Admission Date • Category System: ndhm-supportinginfo-category
Technical Implementation Guidelines	• The file limit for each document should be 2 MB as the maximum size for the whole bundle can only be 20 MB. • Documents need to be double encrypted in Base64 format. There can only be one document linked per item (for e.g., if one item has multiple documents, they need to be merged into one) • Procedure components should have the same value based on the response from the insurance plan • Documents and questionnaires that are mandatory for pre-authorisation will come for the Coverage Eligibility 'auth requirement' response • Supporting documents need to be sent in a fully structured FHIR format. This will have to be aligned with the Health Information (HI) Types defined under ABDM.
Functional points to note	• Pre-authorisation cannot be raised more than one day in advance • Integrators should ideally call coverage eligibility every time a new/additional treatment (should be treated as a validation) is added to ensure it is within the limits before submitting a pre-authorisation
Questionnaire Responses as part of Pre-authorisation Bundle	• For submitting questionnaire responses in a pre-auth/claim request: it needs to be configured in 'Supporting Info' with category INF, code AT and reference value
APIs to be called	/v1/preauth/submit
API to be Implemented	/v1/preauth/on_submit
Use case description	Submit the Preauthorization request from the Provider end.

Protocol status	request.initiated
Request Payload	Encrypted payload of ClaimBundle
Validations	Payload should be validated against the profiles published by NRCES.
The entity to implement the API is called	Payer
Callback API logic	Callback API should be implemented by provider systems. It should accept the payload in two forms, and it will be derived based on the “type” param of the response. 1. Encrypted format (As per RFC7516) of the payload ClaimResponseBundle and decrypted using the private key of the provider entity. Encrypted payload comes as a response only when the payer processes the request and responds when everything is validated and verified. 2. ProtocolResponse: ProtocolResponse comes as response only when the payer is not able to process the request due to the payload is invalid or not able to decrypt or any protocol errors.
Other reference documents	● Checklist of mandatory fields for a successful pre-auth bundle ● Adjudication codes for PMJAY ● NHCX Preauthorisation API ● Claim - FHIR Implementation Guide for ABDM v6.5.0 ● ClaimResponse - FHIR Implementation Guide for ABDM v6.5.0 ● NHCX Workflow IDs

Pre-Authorisation Scenarios

8.4.1 Auto-approval / manual adjudication

Description	The bundle structure for both of these remains the same. The request will get auto-approved/rejected if it adheres to certain parameters. If it does not, it will be manually responded to. If the case is not responded to within the stipulated time, it will automatically get approved. A case will be auto-approved only if the selected procedure(s) are eligible for auto approval. For auto approval to apply, the pre-authorization request must be the first pre-auth for the case, and all requested procedures must be eligible for auto approval. Only when both conditions are met will the case be auto-approved. Separately, cases may also be auto-approved under the TAT-based approval rule. If the policy is eligible for TAT approval and no action is taken within the defined TAT timeframe, the case will be automatically approved.
--------------------	--

How to raise	Standard pre-authorisation format with the relevant workflow ID
---------------------	---

8.4.2 Resubmission/Reprocess

Description	Once a base pre-auth request already exists in the system, a resubmission can be raised. Resubmission will nullify all the previous instances and the payer system will consider this request as the base request.
Example scenario	If there is an approved (or rejected) pre-authorisation or enhancement case, but this needs to be revised for a higher amount or different package, resubmission has to be initiated.
How to raise	Same as pre-authorisation request bundle but with workflow ID for Pre-auth reprocess

8.4.3 Enhancement

Description	Once there is an already approved pre-auth request, an enhancement request needs to be raised to extend the procedure or add new procedures.
Functional points to note	- Multiple enhancements can be raised (unlimited) until discharge as long as it is within the limit (after initial pre-authorisation). Enhancement requests can only be sent after any previous request (pre-auth or resubmission) has been closed.
How to raise	Following items need to be in the bundle: - Workflow ID for enhancement, - Already approved treatments and - Treatments for approval

8.4.4 Query updation

Description	If there is a query in the pre-authorisation submitted, this will be noted in the pre-auth response bundle which has an item-wise adjudication field. Providers should be able to read this query and respond to the same.
How to raise	Once a query has been received, a pre-auth query response (with relevant workflow ID) has to be initiated, and not a resubmission. The bundle structure will be the same.
Sample response	To test this, currently it has to be flagged to the TCS team along with correlation ID and number. They will respond with a dummy query that the integrator can then respond to.

8.4.5 Cancellation

Description	The complete pre-authorisation request gets cancelled. Can be raised if
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	there is an active pre-auth or if there is a pre-auth with decision pending at the payer end.
How to raise	Task Reprocess Request Bundle should be used for pre-auth cancellation (format can be found here) The task component should have: • type of input parameter code which is the intimation number in pre-auth (and claim number in claim), • code for this task is 'cancel' (http://terminology.hl7.org/CodeSystem/financialtaskcode), • reason code based on whatever is used in the system, • remarks can be set in the disposition parameter

8.5. Claim

8.5.0 Structured Data Exchange

Supporting documents need to be sent in a fully structured FHIR format. This will have to be aligned with the Health Information (HI) Types defined under ABDM.

- When **submitting structured clinical data** as part of a **Preauthorization or Claim**, the information must be shared through the **SupportingInfo → DocumentReference** resource.
- Within the **DocumentReference.content.attachment** element, the **attachment.data** field should contain a **Base64-encoded FHIR Bundle**. This bundle should include the relevant structured FHIR resources (for example, **DiagnosticReport**, **DischargeSummary**, **WellnessRecord**, and any associated resources) required to support the preauth or claim.
- This differs from **unstructured submissions**, where the attachment typically contains a **Base64-encoded PDF or JPG document**.
- For **structured submissions**:
 - The attachment represents a **FHIR Bundle encoded in Base64**, not a static document.
 - The bundle contains the necessary **FHIR resources describing the clinical information** relevant to the claim or preauthorization.
The **DocumentReference.content.attachment.contentType** should be set to either **application/json** or **application/fhir+json**.
- For structured data, category code of supporting info needs to be among DIA, HDS, CD, INF and the value needs to be reference.
- For unstructured data, the category code of supporting info needs to be among POI, POA, DOB, DEF, FIR, ATT and the value needs to be attachment.
- Reference documents:
 - [ABDM Sandbox Health Record Formats](#)
 - [Understanding the Structure of ABDM HI Types its Creation and Validation](#)

- [FHIR Profiles for ABDM Health Data Interchange](#)

Key input parameters	All documents to be submitted and questionnaires for Claim will be found in the Insurance Plan response. The 'Coverage Eligibility API' with purpose 'auth requirements' will have those required only for Pre-auth. The remaining documents are mandatory for Claim. The following dates need to be sent mandatorily in the NRCeS prescribed date-time format with the correct codes: Registration Date • Code: EDT • Display: EncounterDateTime • Code System: ndhm-supportinginfo-code • Category Code: OTH • Category Display: Other • Category System: ndhm-supportinginfo-category Admission Date • Code: ADDD • Display: Admission date – Discharge date • Code System: ndhm-supportinginfo-code • Category Code: ADMD • Category Display: Admission Date • Category System: ndhm-supportinginfo-category Surgery Date • Code: ADDD • Display: Admission date – Discharge date • Code System: ndhm-supportinginfo-code • Category Code: SURD • Category Display: Surgery Date • Category System: ndhm-supportinginfo-category Discharge Date • Code: ADDD • Display: Admission date – Discharge date • Code System: ndhm-supportinginfo-code • Category Code: DSCHD • Category Display: Discharge Date • Category System: ndhm-supportinginfo-category Discharge Status • Code: DTH
-----------------------------	--

	• Display: Discharge To Home (Discharge disposition status) • Code System: ndhm-supportinginfo-code • Category Code: DIS • Category Display: Discharge status and discharge to location detail • Category System: ndhm-supportinginfo-category
Technical Implementation Guidelines	• Supporting documents need to be sent in a fully structured FHIR format. This will have to be aligned with the Health Information (HI) Types defined under ABDM. ○ For structured data, category code of supporting info needs to be among DIA, HDS, CD, INF.. and the value needs to be reference.. ○ For unstructured data, the category code of supporting info needs to be among POI, POA, DOB, DEF, FIR, ATT.. and the value needs to be attachment.
Functional points to note	• A claim cannot be cancelled. • The PMJAY system does not have a separate discharge workflow. When a claim is raised, the system implicitly assumes that the patient has been discharged. Accordingly, discharge details must be included as part of the claim request itself. Once the pre-authorisation is approved and the treatment is completed, the patient may be discharged and the claim can be submitted.
Questionnaire Responses as part of Claim Request Bundle	• For submitting questionnaire responses in a pre-auth/claim request: it needs to be configured in 'Supporting Info' with category INF, code AT and reference value
APIs to be called	/v1/claim/submit (for claim submission) /v1/claim/on_submit (for claim response)
API to be Implemented	/v1/claim/on_submit (for claim submission) /v1/claim/submit (for claim response)
Use case description	Submit the Claim from the Provider end to the NHCX.
Protocol status	Payer
Request Payload	request.initiated
Validations	Encrypted payload of ClaimBundle

The entity to implement the API is called	Payload should be validated against the profiles published by NRCES.
Callback API logic	Callback API should be implemented by provider systems. It should accept the payload in two forms, and it will be derived based on the “type” parameter of the response. 1. Encrypted format (As per RFC7516) of the payload ClaimResponseBundle and decrypt using the private key of the provider entity. Encrypted payload comes as a response only when the payer has processed the request and responded, and everything is validated.
Other reference documents	● Checklist of mandatory fields for a successful pre-auth bundle ● Adjudication codes for PMJAY ● NHCX Claim API ● Claim - FHIR Implementation Guide for ABDM v6.5.0 ● ClaimResponse - FHIR Implementation Guide for ABDM v6.5.0 ● Reference documents for structured data: ○ ABDM Sandbox Health Record Formats ○ Understanding the Structure of ABDM HI Types its Creation and Validation ○ FHIR Profiles for ABDM Health Data Interchange

8.5.1 Manual adjudication (approval and rejection)

Description	The bundle structure for both of these remains the same. The request will get approved or rejected on a case-by-case basis.
How to raise	Standard claim format

8.5.2 Query updation

Description	The bundle structure for both of these remains the same. The request will get approved or rejected on a case-by-case basis.
How to raise	Once a query has been received, a claim query response (with workflow ID for claim query response) has to be initiated. The bundle structure will be the same as the Claim Bundle.
Sample response	To test this, currently it has to be flagged to the TCS team along with correlation ID and number. They will respond with a dummy query that the integrator can then respond to.

8.5.3 Erroneous flow

Description	This is a use case for if the claim approved amount is less than the claimed amount . For the remaining amount that has been requested, it will go through the different levels of processing.
How to raise	They need to initiate task request with workflow ID for arbitration request and the following details: • Claim number • Difference amount for which it is being raised

8.5.4 Reprocess/Resubmit

Description	This is a use case for if the claim is rejected and then has to be resubmitted.
How to raise	Once a pre-auth has been approved, a claim has been raised and gets rejected, the claim has to be resubmitted. The same bundle structure but with workflow ID used for reprocess.
Sample request and response	https://www.nrces.in/ndhm/fhir/r4/StructureDefinition-Task.html

8.5.5 Payment Notice

Description	Once a claim has been approved by the payer and the payment has been made, a payment notice is sent to the provider system. Payment notice will give overall consolidated status: • Cleared - amount has been initiated • Paid - amount has been received • Rejected - amount has been initiated but failed due to any server issues • Adjusted - any balance being adjusted Payment reconciliation will give a breakup of payment for that case.
How to raise	The provider system should have the capability to accept and acknowledge the same.

9. APPENDIX

Glossary

Term	Meaning
ABDM	Ayushman Bharat Digital Mission
NHCX	National Health Claims Exchange
PMJAY	Pradhan Mantri Jan Arogya Yojana
TPA	Third Party Administrator
FHIR	Fast Healthcare Interoperability Resources
HMIS	Hospital Management Information System
TMS	Transaction Management System